

Aluminum 2024-T6

Categories: [Metal](#); [Nonferrous Metal](#); [Aluminum Alloy](#); [2000 Series Aluminum Alloy](#)

Material Notes: Weldability = C; Stress Corrosion Cracking Resistance = B; General Corrosion Resistance = D (A = best; E = worst). Good machinability and surface finish capabilities. A high strength material of adequate workability. Has largely superseded 2017 for structural applications.

Uses: Aircraft fittings, gears and shafts, bolts, clock parts, computer parts, couplings, fuse parts, hydraulic valve bodies, missile parts, munitions, nuts, pistons, rectifier parts, worm gears, fastening devices, veterinary and orthopedic equipment, structures.

Some data provided by Alcoa.

Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

Composition Notes:

A Zr + Ti limit of 0.20 percent maximum may be used with this alloy designation for extruded and forged products only, but only when the supplier and the purchaser have mutually agreed.

Composition information provided by the Aluminum Association and is not for design.

Key Words: Aluminium 2024-T6; UNS A92024; ISO AlCu4Mg1; AA2024-T6; Al2024-T6

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	2.78 g/cc	0.100 lb/in ³	AA; Typical
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	125	125	500 kg load/10 mm ball
Hardness, Knoop	157	157	Estimated from Brinell
Hardness, Rockwell A	48	48	Estimated from Brinell
Hardness, Rockwell B	78	78	Estimated from Brinell
Hardness, Vickers	142	142	Estimated from Brinell
Tensile Strength, Ultimate	>= 415 MPa	>= 60200 psi	Wire, rod, and bar (rolled or cold finished); T62
	>= 425 MPa	>= 61600 psi	Wire, rod, and bar (rolled or cold finished)
	>= 427 MPa	>= 61900 psi	
	34.0 MPa @Temperature 371 °C	4930 psi @Temperature 700 °F	
	52.0 MPa @Temperature 316 °C	7540 psi @Temperature 601 °F	
	76.0 MPa @Temperature 260 °C	11000 psi @Temperature 500 °F	
	179 MPa @Temperature 204 °C	26000 psi @Temperature 399 °F	
	310 MPa @Temperature 149 °C	45000 psi @Temperature 300 °F	
	448 MPa @Temperature 100 °C	65000 psi @Temperature 212 °F	
	476 MPa @Temperature 24.0 °C	69000 psi @Temperature 75.2 °F	
	483 MPa @Temperature -28.0 °C	70100 psi @Temperature -18.4 °F	
	496 MPa @Temperature -80.0 °C	71900 psi @Temperature -112 °F	

	579 MPa @Temperature -196 °C	84000 psi @Temperature -321 °F	
	>= 435 MPa @Thickness 12.7 - 76.2 mm	>= 63100 psi @Thickness 0.500 - 3.00 in	Sheet and plate; T62
	>= 440 MPa @Thickness 0.254 - 12.7 mm	>= 63800 psi @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
Tensile Strength, Yield	>= 315 MPa	>= 45700 psi	Wire, rod, and bar (rolled or cold finished); T62
	>= 345 MPa	>= 50000 psi	
	>= 345 MPa @Thickness 0.254 - 12.7 mm	>= 50000 psi @Thickness 0.0100 - 0.499 in	Wire, rod, and bar (rolled or cold finished) Sheet and plate; T62
	>= 345 MPa @Thickness 12.7 - 76.2 mm	>= 50000 psi @Thickness 0.500 - 3.00 in	Sheet and plate; T62
	28.0 MPa @Strain 0.2 %, Temperature 371 °C	4060 psi @Strain 0.2 %, Temperature 700 °F	
	41.0 MPa @Strain 0.2 %, Temperature 316 °C	5950 psi @Strain 0.2 %, Temperature 601 °F	
	62.0 MPa @Strain 0.2 %, Temperature 260 °C	8990 psi @Strain 0.2 %, Temperature 500 °F	
	131 MPa @Strain 0.2 %, Temperature 204 °C	19000 psi @Strain 0.2 %, Temperature 399 °F	
	248 MPa @Strain 0.2 %, Temperature 149 °C	36000 psi @Strain 0.2 %, Temperature 300 °F	
	372 MPa @Strain 0.2 %, Temperature 100 °C	54000 psi @Strain 0.2 %, Temperature 212 °F	
	393 MPa @Strain 0.2 %, Temperature 24.0 °C	57000 psi @Strain 0.2 %, Temperature 75.2 °F	
	400 MPa @Strain 0.2 %, Temperature -28.0 °C	58000 psi @Strain 0.2 %, Temperature -18.4 °F	
	407 MPa @Strain 0.2 %, Temperature -80.0 °C	59000 psi @Strain 0.2 %, Temperature -112 °F	
	469 MPa @Strain 0.2 %, Temperature -196 °C	68000 psi @Strain 0.2 %, Temperature -321 °F	
Elongation at Break	5.0 %	5.0 %	
	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished)
	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished); T62
	10 % @Temperature -80.0 °C	10 % @Temperature -112 °F	
	10 % @Temperature -28.0 °C	10 % @Temperature -18.4 °F	
	10 % @Temperature 24.0 °C	10 % @Temperature 75.2 °F	
	10 % @Temperature 100 °C	10 % @Temperature 212 °F	
	11 % @Temperature -196 °C	11 % @Temperature -321 °F	
	17 % @Temperature 149 °C	17 % @Temperature 300 °F	
	27 % @Temperature 204 °C	27 % @Temperature 399 °F	
	55 % @Temperature 260 °C	55 % @Temperature 500 °F	

	75 % @Temperature 316 °C	75 % @Temperature 601 °F	
	100 % @Temperature 371 °C	100 % @Temperature 700 °F	
	>= 5.0 % @Thickness 0.254 - 12.7 mm	>= 5.0 % @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
	>= 5.0 % @Thickness 12.7 - 76.2 mm	>= 5.0 % @Thickness 0.500 - 3.00 in	Sheet and plate; T62
Modulus of Elasticity	72.4 GPa	10500 ksi	Estimated from other heat treatments.
Poissons Ratio	0.33	0.33	Estimated from other heat treatments.
Fatigue Strength	124 MPa @# of Cycles 5.00e+8	18000 psi @# of Cycles 5.00e+8	completely reversed; R. R. Moore Machine and specimen.
Machinability	70 %	70 %	0-100 Scale (A=90; B=70; C=50; D=30; E=10)
Shear Modulus	27.0 GPa	3920 ksi	Estimated from similar Al alloys.
Shear Strength	283 MPa	41000 psi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000449 ohm-cm @Temperature 20.0 °C	0.00000449 ohm-cm @Temperature 68.0 °F	AA; Typical

Thermal Properties	Metric	English	Comments
CTE, linear 	21.1 µm/m-°C @Temperature -50.0 - 20.0 °C	11.7 µin/in-°F @Temperature -58.0 - 68.0 °F	
	22.9 µm/m-°C @Temperature 20.0 - 100 °C	12.7 µin/in-°F @Temperature 68.0 - 212 °F	
	23.2 µm/m-°C @Temperature 20.0 - 100 °C	12.9 µin/in-°F @Temperature 68.0 - 212 °F	AA; Typical; average over range
	23.8 µm/m-°C @Temperature 20.0 - 200 °C	13.2 µin/in-°F @Temperature 68.0 - 392 °F	
	24.7 µm/m-°C @Temperature 20.0 - 300 °C	13.7 µin/in-°F @Temperature 68.0 - 572 °F	
Specific Heat Capacity	0.875 J/g-°C	0.209 BTU/lb-°F	
Thermal Conductivity	151 W/m-K	1050 BTU-in/hr-ft²-°F	AA; Typical at 77°F
Melting Point	502 - 638 °C	935 - 1180 °F	AA; Typical range based on typical composition for wrought products >= 1/4 in. thickness. Eutectic melting is not eliminated by homogenization.
Solidus	502 °C	935 °F	AA; Typical
Liquidus	638 °C	1180 °F	AA; Typical

Processing Properties	Metric	English	Comments
Annealing Temperature	413 °C	775 °F	
Solution Temperature	493 °C	919 °F	
Aging Temperature	191 °C	375 °F	8 to 16 hr at temperature

Component Elements Properties	Metric	English	Comments
Aluminum, Al	90.7 - 94.7 %	90.7 - 94.7 %	As remainder
Chromium, Cr	<= 0.10 %	<= 0.10 %	

Copper, Cu	3.8 - 4.9 %	3.8 - 4.9 %
Iron, Fe	<= 0.50 %	<= 0.50 %
Magnesium, Mg	1.2 - 1.8 %	1.2 - 1.8 %
Manganese, Mn	0.30 - 0.90 %	0.30 - 0.90 %
Other, each	<= 0.05 %	<= 0.05 %
Other, total	<= 0.15 %	<= 0.15 %
Silicon, Si	<= 0.50 %	<= 0.50 %
Titanium, Ti	<= 0.15 %	<= 0.15 %
Zinc, Zn	<= 0.25 %	<= 0.25 %

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.